

SAVONLINNA PUNKAHARJU, Finland

WMO: 027780

Lat: 61.8019N

Lon: 29.3153E

Elev: 261

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.4	(c) -9.2	(d) -20.4	(e) 1.8	(f) -15.3	(g) -13.3	(h) 2.7	(i) -9.0	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.4	(c) 81.1	(d) 66.3	(e) 77.3	(f) 64.7	(g) 73.8	(h) 62.6	(i) 68.9	(j) 77.3	(k) 66.8	(l) 74.2	(m) 64.7	(n) 71.8	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	96.8	72.3	64.0	90.0	70.6	61.7	83.1	68.0	33.2	77.4	31.5	74.2	29.9	71.9	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -17.2	(f) 83.8	(g) 8.7	(h) 4.6	(i) -23.5	(j) 87.1	(k) -28.5	(l) 89.8	(m) -33.4	(n) 92.3	(o) -39.7	(p) 95.6		
(4) N/A	(5) N/A	(6) N/A	(e) -17.5	(f) 71.0	(g) 8.6	(h) 3.2	(i) -23.7	(j) 73.2	(k) -28.7	(l) 75.1	(m) -33.5	(n) 76.9	(o) -39.7	(p) 79.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.9	15.7	19.3	26.2	37.4	50.1	57.5	63.3	60.3	51.7	39.6	32.3	24.3		
	DBStd	17.99	13.49	12.34	8.80	6.37	7.82	6.76	6.20	5.55	5.69	6.19	7.23	10.83		
	HDD50	4853	1064	860	737	380	97	14	0	2	48	325	530	796		
	HDD65	9246	1529	1280	1202	829	465	239	107	167	399	788	980	1261		
	CDD50	1174	0	0	0	1	99	239	414	320	99	2	0	0		
	CDD65	91	0	0	0	0	2	14	55	20	0	0	0	0		
	CDH74	730	0	0	0	0	46	134	415	135	0	0	0	0		
	CDH80	139	0	0	0	0	3	21	92	23	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	24.3	1.8	1.5	1.3	1.1	1.6	2.4	2.7	3.0	2.3	2.4	2.2	2.0		
(15) Precipitation	PrecMax	29.3	2.9	2.9	2.3	2.9	4.5	3.9	5.8	5.3	6.5	5.1	4.1	3.3		
	PrecMin	18.8	0.5	0.2	0.2	0.3	0.3	0.7	0.9	0.7	0.8	0.6	0.4	0.8		
	PrecStd	3.0	0.7	0.7	0.6	0.6	1.0	0.8	1.3	1.5	1.3	1.1	0.9	0.7		
	PrecStd	3.0	0.7	0.7	0.6	0.6	1.0	0.8	1.3	1.5	1.3	1.1	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	40.3	48.5	63.7	79.1	82.8	86.6	83.1	68.7	55.8	49.6	43.5		
		MCWB	36.1	36.3	39.1	49.5	60.4	63.9	69.7	68.8	60.6	52.7	47.4	42.2		
	2%	DB	35.8	36.8	42.9	57.0	73.9	77.7	82.5	77.7	64.9	52.5	45.7	39.6		
		MCWB	34.9	35.0	37.0	46.0	57.6	63.7	68.1	65.4	58.6	50.1	44.4	38.1		
	5%	DB	33.8	34.9	39.3	52.4	69.1	73.8	78.9	73.5	62.3	50.8	43.6	37.2		
		MCWB	33.0	33.7	34.9	42.8	55.3	61.2	66.8	63.3	57.3	48.7	42.5	36.2		
	10%	DB	32.4	33.5	36.8	48.4	64.5	70.1	74.8	70.1	60.2	48.8	41.2	35.1		
		MCWB	31.7	32.8	33.8	41.0	52.4	59.1	64.3	62.1	56.2	46.8	40.3	34.4		
	0.4%	WB	36.7	37.3	40.4	50.8	63.9	69.1	72.4	71.3	62.0	53.0	47.6	42.3		
		MCDB	37.1	39.2	45.7	60.2	74.9	78.3	81.6	81.0	66.3	54.3	48.8	43.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.0	35.4	37.7	47.4	59.7	65.9	70.0	67.1	60.2	51.1	44.4	38.6		
		MCDB	35.6	36.8	41.4	55.6	69.0	73.5	79.5	74.1	63.4	52.1	45.3	39.5		
	5%	WB	33.3	34.0	35.9	44.4	57.0	62.6	67.9	64.9	58.5	49.1	42.7	36.3		
		MCDB	33.9	34.8	38.5	50.9	66.6	70.8	75.5	71.6	61.6	50.2	43.7	37.0		
	10%	WB	32.0	32.9	34.4	41.9	54.2	60.2	66.2	63.0	56.9	46.9	40.5	34.4		
		MCDB	32.7	33.6	36.3	47.1	63.2	68.1	73.2	68.6	59.3	48.5	41.2	35.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	10.9	14.2	16.4	18.5	16.4	15.4	14.1	11.0	8.4	6.0	7.2		
		MCDBR	5.8	9.1	16.2	23.3	25.9	22.2	21.1	19.7	14.7	10.7	8.1	6.8		
		MCWBR	5.8	7.9	11.2	13.7	12.3	10.2	8.6	9.1	9.0	8.5	7.7	6.5		
	5% WB	MCDBR	5.6	8.3	13.5	20.4	21.7	20.0	18.6	17.3	12.8	9.4	7.9	6.5		
		MCWBR	5.7	7.4	10.0	12.5	11.9	10.3	8.7	8.7	8.6	8.4	7.6	6.5		
			(d) 10.5	(e) 10.9	(f) 14.2	(g) 16.4	(h) 18.5	(i) 16.4	(j) 15.4	(k) 14.1	(l) 11.0	(m) 8.4	(n) 6.0	(o) 7.2		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.294	0.320	0.359	0.355	0.351	0.376	0.361	0.338	0.319	0.265	N/A		
	taud		2.039	2.116	2.193	2.337	2.444	2.486	2.431	2.457	2.510	2.401	2.127	N/A		
	Ebn at Noon		137	207	248	263	275	278	266	261	247	204	129	N/A		
	Edh at Noon		12	23	30	32	31	30	32	29	23	17	11	N/A		
			(d) 137	(e) 207	(f) 248	(g) 263	(h) 275	(i) 278	(j) 266	(k) 261	(l) 247	(m) 204	(n) 129	(o) N/A		
(44) All-Sky Solar Radiation	RadAvg		71	248	716	1157	1569	1680	1623	1247	711	282	75	35		
	RadStd		11	61	102	130	184	102	154	149	81	41	12	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page